

1. **Number of Questions:** In the final examination, 12 questions of 5 marks each should be asked.
2. **Marks distribution as follows:**

| Chapter 1.<br>Descriptive Statistics and Basic<br>Probability (6 hours)<br>2 questions of 5 marks each                                                                                                                 | Marks | No. of<br>question | Remarks on question pattern                                                                                                                                                                                      |
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| 1.1 Introduction to statistics and its<br>importance in engineering<br>1.2 Measure of central tendency and<br>measure of variation<br>1.3 Graphical representation of data:<br>Histograms, box plots and scatter plots | 5     | 1                  | <ul style="list-style-type: none"> <li>• Only quartiles, percentile not required</li> <li>• Can use calculator for raw data to calculate mean, median, but students should write theoretical formula.</li> </ul> |
| 1.4 Basic probability concepts, additive law,<br>multiplicative law<br>1.5 Conditional probability and Bayes'<br>theorem                                                                                               | 5     | 1                  | <ul style="list-style-type: none"> <li>• Proof of theorem not required.</li> <li>• Application in engineering field can be asked</li> </ul>                                                                      |

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| theorem                                                                                                                   |              |                        |                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>Chapter 2: Probability Distributions and Sampling Distribution (14 hours)</b><br><b>3 questions of 5 marks each</b>    | <b>Marks</b> | <b>No. of question</b> | <b>Remarks</b>                                                                                                                                                                                                                                                                                                                                                                                                                                |
| 2.1 Random variables: Discrete and continuous<br>2.2 Expectation and variance of discrete and continuous random variables | 5            | 1                      | <ul style="list-style-type: none"> <li>• Attempt any three questions.</li> <li>• One question is asked in <b>OR</b> option within the chapter.</li> <li>• For example, choice can be between Binomial and Poisson, Normal distribution and Binomial distribution etc.</li> <li>• Theory + Numerical can be asked in a question (but not only theory)</li> <li>• Marks distribution of Theory + Numerical will be either 2+3 or 1+4</li> </ul> |
| 2.3 Discrete probability distributions: Binomial, Poisson, negative Binomial                                              | 5            | 1                      |                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 2.4 Continuous probability distributions: Normal, Gamma, Chi-Square                                                       | 5            | 1                      |                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 2.5 Population and sample<br>2.6 Sampling distribution of mean and proportion<br>2.7 Central limit theorem                | 5            | 1                      |                                                                                                                                                                                                                                                                                                                                                                                                                                               |

| <b>Chapter 3: Statistical Inference (14 hours)<br/>4 questions of 5 marks each</b>                      | <b>Marks</b> | <b>No. of<br/>question</b> | <b>Remarks</b>                                                                                                                                                                                                                                                                                                                                       |
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| 3.1 Point estimations and properties of estimators<br>3.2 Confidence intervals for mean and proportions | 5            | 1                          | <ul style="list-style-type: none"> <li>• Property of good estimator (concept only)</li> <li>• To find sample size (n)</li> </ul>                                                                                                                                                                                                                     |
| 3.3 Hypothesis testing, parametric and non-parametric tests, procedure of hypothesis                    | 5            | 1                          | <ul style="list-style-type: none"> <li>• Theory Question only, test procedure of any test can be ask</li> </ul>                                                                                                                                                                                                                                      |
| 3.4 Hypothesis testing of mean (Single mean, two means, paired t-test and one-way)                      | 5            | 1                          | <ul style="list-style-type: none"> <li>• As not included in course, do not ask question on Hypothesis test concerning proportion (one sample and two sample) but can be practiced in tutorial class</li> <li>• In testing of equality between means of two samples, include test with non-zero difference also.</li> </ul>                           |
| 3.5 Goodness of fit tests and independence of attributes (Chi-square and Kolmogorov–Smirnov test)       | 5            | 1                          | <ul style="list-style-type: none"> <li>• Goodness of fit for Binomial and Poisson distribution only (test for Normal distribution not required)</li> <li>• Avoid questions that need Yates Correction for Chi-square test of independence</li> <li>• For Chi-square test of independence, Contingency table up to 3×3 only (max catagori)</li> </ul> |

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| 3.1 Point estimations and properties of estimators<br>3.2 Confidence intervals for mean and proportions | 5 | 1 | <ul style="list-style-type: none"> <li>• Property of good estimator (concept only)</li> <li>• To find sample size (n)</li> </ul>                                                                                                                                                                                                                                                                                                                    |
| 3.3 Hypothesis testing, parametric and non-parametric tests, procedure of hypothesis                    | 5 | 1 | <ul style="list-style-type: none"> <li>• Theory Question only, test procedure of any test can be ask</li> </ul>                                                                                                                                                                                                                                                                                                                                     |
| 3.4 Hypothesis testing of mean (Single mean, two means, paired t-test and one-way)                      | 5 | 1 | <ul style="list-style-type: none"> <li>• As not included in course, do not ask question on Hypothesis test concerning proportion (one sample and two sample) but can be practiced in tutorial class</li> <li>• In testing of equality between means of two samples, include test with non-zero difference also.</li> </ul>                                                                                                                          |
| 3.5 Goodness of fit tests and independence of attributes (Chi-square and Kolmogorov–Smirnov test)       | 5 | 1 | <ul style="list-style-type: none"> <li>• Goodness of fit for Binomial and Poisson distribution only (test for Normal distribution not required)</li> <li>• Avoid questions that need Yates Correction for Chi-square test of independence</li> <li>• For Chi-square test of independence, Contingency table up to 3×3 only (max catagori)</li> <li>• Concept only for Kolmogorov–Smirnov test, demonstrate this test in class in laptop.</li> </ul> |

| <b>Chapter 4: Correlation and Regression<br/>(6 hours)<br/>2 questions of 5 marks each</b>                           | <b>Marks</b> | <b>No. of<br/>question</b> | <b>Remarks</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
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| 4.1 Correlation analysis and test of linear correlation                                                              | 5            | 1                          | question from Correlation Simple correlation (linear only), Marks distribution of Theory + Numerical will be either 2+3 or 1+4                                                                                                                                                                                                                                                                                                                                                                   |
| 4.2 Simple regression analysis, the concept of explained, unexplained, and total<br>4.3 Multiple regression analysis | 5            | 1                          | <ul style="list-style-type: none"> <li>• Confidence interval for slope “b” only in tutorial</li> <li>• Marks distribution of Theory + Numerical will be either 2+3 or 1+4</li> <li>• Show explained, unexplained and total variation in graph only</li> <li>• <u>For multiple regression:</u></li> <li>• Give raw data of up to 8 observations,</li> <li>• students can use calculator to find sums, and write it without showing the workouts, but writing the formula is necessary.</li> </ul> |

| <b>Chapter 5: Statistical Quality Control<br/>(5 hours)<br/>1 question of 5 marks</b>                                                         | <b>Marks</b> | <b>No. of<br/>question</b> | <b>Remarks</b>                                                                                                                                                                                                                                                                                                                                        |
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| 5.1 Quality control and its importance in engineering<br>5.2 Control charts for variables (X-bar, R-chart, P-chart)<br>5.3 Six sigma concepts | 5            | 1                          | <ul style="list-style-type: none"> <li>• create control chart for fix sample size only, time with data set up to 5 or mean and range should be provide.</li> <li>• A2, D3 and D4 should be provided,</li> <li>• general introduction and question related with six sigma concept theory can be ask but not necessary, it should 1+4 of 2+3</li> </ul> |

Remarks: